

First-Order ODEs

Zill 2.2–2.5

Separable separate, integrate	$\frac{dy}{dx} = g(x)h(y)$ $\int \frac{dy}{h(y)} = \int g(x) dx + C$
Linear integrating factor μ	$y' + P(x)y = Q(x), \quad \mu(x) = e^{\int P(x) dx}$ $y = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x) dx + C \right]$
Exact potential fn. f	$M dx + N dy = 0 \quad \text{exact iff} \quad M_y = N_x$ $f = \int M dx + g(y), \quad \text{solve } g \text{ from } f_y = N$ $f(x, y) = C$
Non-exact restore $M_y = N_x$	$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx} \quad \text{if } x \text{ only}$ $\mu(y) = e^{\int \frac{N_x - M_y}{M} dy} \quad \text{if } y \text{ only}$
Homogeneous $\rightarrow$ separable in v, x	$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ $y = vx, \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$
Bernoulli $\rightarrow$ linear in w	$y' + P(x)y = Q(x)y^n, \quad w = y^{1-n}$ $\frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$

Linear Higher-Order ODEs

Zill 3.1–3.6

General solution	$y = y_c + y_p$ y_c solves the homogeneous eqn; y_p is any one particular solution.
Reduction of order 2nd sol. from known y_1	$y'' + P(x)y' + Q(x)y = 0$ $y_2 = y_1(x) \int \frac{e^{-\int P dx}}{[y_1(x)]^2} dx$

	Const. coeff. $ay'' + by' + cy = 0$	Cauchy–Euler , $x > 0$ $ax^2y'' + bxy' + cy = 0$
	$y = e^{mx}$: $am^2 + bm + c = 0$	$y = x^m$: $am(m-1) + bm + c = 0$
Distinct	$c_1e^{m_1x} + c_2e^{m_2x}$	$c_1x^{m_1} + c_2x^{m_2}$
Repeated	$c_1e^{mx} + c_2xe^{mx}$	$c_1x^m + c_2x^m \ln x$
$\alpha \pm i\beta$	$e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$	$x^\alpha[c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$
nth order, repeated root of mult. k	real m : $e^{mx}, xe^{mx}, \dots, x^{k-1}e^{mx}$ pair $\alpha \pm i\beta$: $x^je^{\alpha x} \cos \beta x, x^je^{\alpha x} \sin \beta x$ $j = 0, \dots, k-1$	

$g(x)$	trial y_p
constant	A
poly., degree n	$A_nx^n + \dots + A_1x + A_0$
e^{ax}	Ae^{ax}
$\cos \beta x$ or $\sin \beta x$	$A \cos \beta x + B \sin \beta x$
product of these	product of the forms
A term of the guess already in y_c ? Multiply the guess by x (or x^2).	

Variation of parameters $y'' + Py' + Qy = f(x)$	$y_c = c_1y_1 + c_2y_2, \quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ $u_1' = \frac{-y_2f(x)}{W}, \quad u_2' = \frac{y_1f(x)}{W}$ $y_p = u_1y_1 + u_2y_2$
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$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) \, dt$$

$t \geq 0$; converges for large s if f is piecewise continuous and of exponential order.

$f(t)$	$F(s)$	$f(t)$	$F(s)$
1	$\frac{1}{s}$	t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$	$e^{at}t^n$	$\frac{n!}{(s-a)^{n+1}}$
$\sin kt$	$\frac{k}{s^2+k^2}$	$\cos kt$	$\frac{s}{s^2+k^2}$
$\sinh kt$	$\frac{k}{s^2-k^2}$	$\cosh kt$	$\frac{s}{s^2-k^2}$
$t \sin kt$	$\frac{2ks}{(s^2+k^2)^2}$	$t \cos kt$	$\frac{s^2-k^2}{(s^2+k^2)^2}$
$e^{at} \sin kt$	$\frac{k}{(s-a)^2+k^2}$	$e^{at} \cos kt$	$\frac{s-a}{(s-a)^2+k^2}$
$\mathcal{U}(t-a)$	$\frac{e^{-as}}{s}$	$\delta(t-a)$	e^{-as}

Valid for $s > 0$; $s > a$ when e^{at} appears; $s > |k|$ for $\sinh$, $\cosh$; $a \geq 0$ for $\mathcal{U}$, δ ; $n = 1, 2, 3, \dots$

Linearity also for $\mathcal{L}^{-1}$	$\mathcal{L}\{af + bg\} = aF(s) + bG(s)$ $\mathcal{L}^{-1}\{aF + bG\} = a\mathcal{L}^{-1}\{F\} + b\mathcal{L}^{-1}\{G\}$ Inverse: partial fractions, then invert term by term.
Derivatives brings in $y(0)$, $y'(0)$	$\mathcal{L}\{f'\} = sF(s) - f(0)$ $\mathcal{L}\{f''\} = s^2F(s) - sf(0) - f'(0)$ $\mathcal{L}\{f^{(n)}\} = s^n F - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$

Integral	$\mathcal{L}\left\{\int_0^t f(\tau) \, d\tau\right\} = \frac{F(s)}{s}$
1st shift replace $s \rightarrow s - a$	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
Unit step	$\mathcal{U}(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$
2nd shift delay by a	$\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as}F(s)$ $\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}$
Dirac delta unit impulse	$\int_0^\infty f(t)\delta(t-a) \, dt = f(a)$ $\mathcal{L}\{\delta(t-a)\} = e^{-as}, \quad \mathcal{L}\{\delta(t)\} = 1$
Convolution product $F \cdot G$	$(f * g)(t) = \int_0^t f(\tau)g(t-\tau) \, d\tau$ $\mathcal{L}\{f * g\} = F(s)G(s)$
Periodic $f(t+T) = f(t)$	$\mathcal{L}\{f\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) \, dt$

SOLVING AN IVP BY LAPLACE

- 1 Transform both sides with $\mathcal{L}\{y^{(n)}\} \rightarrow$ algebraic eqn in $Y(s)$.
- 2 Solve algebraically for $Y(s)$.
- 3 Partial fractions to match table entries; shift theorems for $e^{\pm as}$ factors; convolution for a product $F(s)G(s)$.
- 4 Invert term by term: $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.